

ZHOB, Pakistan

WMO: 416200

Lat: 31.358N

Lon: 69.464E

Elev: 4650

StdP: 12.39

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.9	(c) 32.7	(d) -1.7	(e) 6.0	(f) 58.4	(g) 3.4	(h) 7.8	(i) 59.9	(j) 16.6	(k) 53.8	(l) 12.2	(m) 50.0	(n) 4.5	(o) 180	(p) 0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 20.8	(c) 100.9	(d) 65.7	(e) 98.9	(f) 65.4	(g) 97.0	(h) 65.2	(i) 76.1	(j) 88.7	(k) 74.2	(l) 87.2	(m) 72.6	(n) 86.3	(o) 7.1	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.8	145.1	84.6	70.6	134.1	82.9	68.7	125.8	81.2	43.5	89.1	41.5	87.6	39.8	86.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	14.3	11.7	26.8	105.4	3.0	1.8	24.6	106.7	22.9	107.7	21.3	108.7	19.1	110.0		
			22.7	80.5	2.6	3.7	20.8	83.2	19.3	85.3	17.8	87.5	15.9	90.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.2	45.2	49.5	59.1	70.7	79.5	85.6	86.1	84.8	80.0	70.5	57.7	48.5
	DBStd	15.71	4.45	6.71	6.84	6.32	5.32	4.52	4.13	4.01	3.71	5.60	5.29	4.73
	HDD50	335	154	83	7	0	0	0	0	0	0	0	5	86
	HDD65	2023	612	435	206	21	0	0	0	0	0	14	223	512
	CDD50	6980	7	68	290	621	915	1069	1120	1077	901	636	238	38
	CDD65	3193	0	0	24	192	450	619	655	612	451	185	5	0
	CDH74	39208	0	15	302	2026	5521	8361	8512	7484	5034	1843	102	8
	CDH80	20141	0	1	58	765	2850	4768	4754	3924	2407	608	6	0
Wind	WSAvg	5.3	5.3	5.0	5.7	6.1	5.9	6.0	5.5	5.6	5.1	4.8	4.4	4.8
Precipitation	PrecAvg	10.4	0.6	0.9	1.6	1.2	0.5	0.6	2.1	1.7	0.5	0.1	0.2	0.4
	PrecMax	16.5	3.1	2.8	4.6	5.3	2.3	3.0	6.4	5.9	2.3	1.0	3.1	2.9
	PrecMin	4.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.0	0.6	0.7	1.2	1.2	0.5	0.7	1.5	1.5	0.6	0.2	0.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	76.0	85.5	93.8	100.0	104.0	103.6	102.0	98.1	91.8	80.6	73.7
		MCWB	45.5	49.9	54.1	58.1	62.6	66.7	67.8	68.0	61.0	59.6	53.0	50.2
	2%	DB	63.8	70.8	81.1	90.6	97.5	101.1	100.5	99.0	95.1	89.1	77.1	68.3
		MCWB	43.9	48.5	53.2	58.2	61.2	64.3	67.2	67.4	63.6	58.2	51.8	47.6
	5%	DB	60.5	67.0	77.4	87.5	95.1	98.8	98.3	96.6	93.0	86.3	74.1	64.9
		MCWB	42.7	47.1	52.2	57.0	60.5	64.3	67.7	67.0	62.8	57.0	50.7	45.5
	10%	DB	57.4	62.9	73.4	84.5	92.6	96.4	95.8	94.5	90.8	83.1	71.0	61.4
		MCWB	41.6	45.9	50.8	56.9	59.8	64.0	68.5	66.9	63.1	56.3	49.4	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	54.9	59.9	67.9	74.1	76.3	79.9	78.8	75.4	67.5	57.5	52.6
		MCDB	60.8	65.2	75.0	85.5	91.9	91.9	93.1	91.6	89.5	79.9	71.4	71.2
	2%	WB	46.1	51.3	56.9	63.1	70.2	73.0	76.5	75.1	71.6	63.9	55.5	49.0
		MCDB	59.0	64.1	72.4	81.0	89.3	91.1	89.0	86.6	85.3	79.7	69.5	64.3
	5%	WB	44.3	49.3	54.5	60.7	66.8	71.1	74.7	73.5	69.8	61.7	53.2	46.8
		MCDB	57.4	63.0	71.3	78.4	84.7	87.6	86.6	85.3	83.2	78.9	67.6	62.1
	10%	WB	42.7	47.3	52.6	59.1	64.1	69.9	72.9	72.2	68.4	59.8	51.4	44.8
		MCDB	55.1	60.7	69.2	76.7	83.1	86.8	86.4	84.8	81.2	76.8	66.6	59.4
Mean Daily Temperature Range	5% DB	MDBR	22.8	21.8	22.4	21.8	22.6	20.8	20.5	19.6	21.7	23.3	23.7	24.4
		MCDBR	28.5	27.5	27.9	25.6	25.1	22.7	22.7	22.1	23.8	24.9	26.9	29.9
		MCWBR	13.9	11.8	9.9	7.6	6.8	6.9	6.9	7.6	8.0	7.3	10.2	14.1
	5% WB	MCDBR	25.6	24.0	24.7	22.4	22.9	20.7	21.1	20.1	22.0	23.1	23.6	27.7
	MCWBR	13.8	12.0	11.3	9.0	10.5	7.8	8.2	7.8	8.7	8.9	10.5	14.5	
Clear-Sky Solar Irradiance	taub	0.329	0.389	0.421	0.470	0.479	0.555	0.635	0.582	0.450	0.393	0.359	0.320	
	taud	2.301	2.115	2.035	1.946	1.931	1.772	1.649	1.757	2.020	2.104	2.194	2.338	
	Ebn at Noon	282	272	272	263	260	240	220	231	259	266	266	279	
	Edh at Noon	34	45	52	59	61	71	80	71	52	45	37	31	
All-Sky Solar Radiation	RadAvg	1148	1347	1733	2067	2322	2274	2118	2026	1930	1657	1286	1113	
	RadStd	82	125	126	108	97	81	75	114	77	47	66	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page